DRAGOON MOUNTAINS GROWING GUIDES

A Hydroponic Primer

Growing Food in the Dragoon Mountains — Cochise County, Arizona

Dragoon Mountains Growing Guides

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This is the orientation guide for the Dragoon Mountains library. Start here if you are new to hydroponic growing in Cochise County; then move sideways into the deeper guides.

- **A — Dragoon Mountains Pepper Growing Guide** — Andean peppers, heritage, and the cornerstone programme
- **B — Bato Bucket Hydroponic Setup** — the recommended build for fruiting crops
- **C — Pepper Hydroponic Guide** — what changes when you take peppers off soil
- **D — A Hydroponic Primer** *(this guide)* — climate, systems, calendar, crops
- **E — Modern Masterblend Guide** — the purchased-nutrient programme
- **F — Hydroponic Nutrients Guide** — locally-sourced nutrient recipes in detail
- **G — pH Up & pH Down Replacements** — acid and alkali from local materials
- **H — Grape Growing Guide** — a companion crop for the same climate

Figure 1 – The Dragoon Mountains rising above the Sulphur Springs Valley, Cochise County, Arizona.

How to use this library

The Dragoon Mountains library is eight short guides that sit beside each other rather than stack on top of each other. Any one of them will get you a harvest; read together, they show you a whole growing system built for one particular sky island in one particular county.

This primer is the shortest path in. It covers the climate you are actually growing in, the water you will actually be dosing, the systems that work at 4,500 to 7,500 feet, a twelve-month calendar, a crop list, and a troubleshooting table. That is enough to plant.

For anything deeper — a nutrient recipe, a specific pepper variety, a bucket build — the primer points you into the companion guide that covers it properly. Nothing important lives in two places.

If you are new, read Chapter 1 (climate) and Chapter 6 (calendar) first, then skim the rest. The Dragoons' seasonal rhythm is the single most important thing to internalise — the monsoon, the long fall, and the cold-tolerant winter window define what grows well here.

1. Know Your Environment

The Dragoon Mountains, in southeastern Cochise County, are a sky island range rising from grassland at 4,500 ft to wooded peaks near 7,500 ft. This geography gives you genuine advantages over low-desert Arizona — cooler summers, reliable monsoon rains, and a rich local mineral geology — but the trade-off is wide day-to-night temperature swings, intense UV, and hard well water.

Climate at a glance

Factor	Condition	What it means for hydroponics
Elevation	4,500 – 7,500 ft	Stronger UV, cooler than low desert, lower air pressure
Summer highs	85 – 95 °F	Manageable — crops set fruit where Phoenix cannot
Night swings	30 – 40 °F diurnal	Stresses plants; insulate or bury the reservoir
Monsoon	July – September	50–70 % humidity; soft rainwater to collect
Annual rainfall	15 – 20 in	Mostly monsoon — plan catchment around it
Winter frost	November – February	Protected structure needed; superb for cold greens

Your key advantages

Cooler summers than low-desert Arizona mean fruit crops don't abort flowers from heat stress. The monsoon delivers soft, low-TDS rainwater that is ideal reservoir source water. The local geology — iron-bearing granite, caliche calcium, sulphur deposits, copper ores — gives you every element needed for a full nutrient profile if you want to brew your own. Oak woodland supplies potassium from ash and chelating tannins from bark. Bat caves throughout the county supply high-nitrogen guano. And the long, warm fall from September to November is the Dragoons' signature growing window — plan your biggest plantings to mature here.

Fall (September – November) is your prime growing season. Set your calendar so the biggest crops finish in this window, not in the heat of July.

2. Water Treatment

Water quality is the single most important variable in hydroponic success in southeastern Arizona. Well water here is typically hard — high in dissolved calcium, magnesium, and especially bicarbonates, which constantly push pH upward. Treating source water before adding nutrients is not optional.

Two meters, non-negotiable

You need an EC/TDS meter to measure what is already in the water (local well water commonly reads 200–500 ppm before any nutrients go in), and a pH meter to measure acidity (local well water often reads 7.5–8.5; your target for most crops is 5.8–6.2).

The bicarbonate scrub

Bicarbonates (HCO_3^-) are the main culprit in hard-water pH instability. Even after you adjust pH down, bicarbonates act as a buffer and push it back up within hours. Neutralise them *first*, before adding nutrients:

Step	Action	What happens
1	Fill the reservoir with source water	Starting pH likely 7.5–8.5
2	Add acid slowly while stirring	Bicarbonates react — the water fizzes, releasing CO_2
3	Wait until fizzing stops completely	Bicarbonates are neutralised
4	Adjust pH to 5.8–6.0	Now pH is stable — it will not bounce back
5	Add nutrients in the correct order	Nutrients dissolve into pre-treated, stable water

Water sources, ranked

Source	TDS	pH	Treatment	Rating
Monsoon rainwater (collected)	< 20 ppm	6.0–6.5	Minimal — just pH fine-tune	★★★ Best
Filtered / RO well water	< 50 ppm	6.5–7.0	Light pH adjustment	★★ Good
Untreated well water	200–500 ppm	7.5–8.5	Full bicarbonate scrub + pH down	■ Usable
Surface / stock-tank water	Variable	Variable	Test first — may contain contaminants	■ Test first

A 500 sq ft roof collects 3,000 + gallons per monsoon season. Store in dark, covered tanks to prevent algae. This is the best water you will ever put through a reservoir in Cochise County.

For the full acid and alkali recipes — prickly-pear vinegar, oak-ash lye, sulphur-burn acid — see **Guide G: pH Up & pH Down Replacements**.

3. System Design & Setup

The right system for the Dragoons addresses three local challenges: intense UV, large temperature swings, and high evaporation. Simplicity and passive resilience matter more here than high-tech solutions.

Recommended systems

System	Best for	Dragoon advantage	Key requirement
Kratky (passive)	Greens, herbs, lettuce	No pump, no electricity — ideal off-grid	Weekly water-level check
DWC (Deep Water Culture)	Tomatoes, peppers, cucumbers	Simple, high yield, easy temp management	Air pump + airstone
Drip / Dutch bucket	Large fruiting crops	Best water efficiency for precious water	Timer + pump
Wicking	Herbs, small plants	Zero electricity, completely passive	Correct media ratio

For the recommended Dutch-bucket build in full — plumbing, reservoir, pump sizing, daily chores — see **Guide B: Bato Bucket Hydroponic Setup**. For the pepper-specific take on all four systems above, see **Guide C: Pepper Hydroponic Guide**.

Reservoir management

Bury or insulate reservoirs — underground tanks stay 10–15 °F cooler, keeping the root zone in the ideal 65–72 °F range. Use dark, opaque, UV-stabilised containers; standard plastic degrades rapidly at Dragoon elevations. Cover every reservoir completely — evaporation in the high desert is extreme, and an uncovered 50-gallon reservoir can lose five gallons a day in summer.

Shade & protection

Use 30–40 % shade cloth for summer growing; high-altitude UV bleaches leaves and heats reservoirs. East-facing orientation gives you morning sun and afternoon shade from the mountains. A simple ramada of native juniper or oak poles with a shade-cloth roof is low-cost and blends with the landscape. A hoophouse extends your season through the frost months.

Growing media

Medium	Best use	Local source?
Perlite	Most systems — excellent drainage, reusable	Purchase
Coarse granite sand	Wicking, drip	✓ Crush local granite — free and abundant
Gravel (washed)	Dutch bucket, large systems	✓ Local creek gravel — wash thoroughly
Coconut coir	Water retention in hot weather	Purchase
Pumice	Drainage + aeration	✓ Volcanic pumice found in SE Arizona

Crushed local granite and washed creek gravel make excellent free media. Rinse, soak in dilute vinegar to remove carbonates, rinse again — ready.

4. Nutrients — short version

The Dragoons and surrounding county contain nearly every element needed to build a complete hydroponic nutrient solution — potassium from oak ash, phosphorus and nitrogen from bat guano, calcium from caliche, magnesium and sulphur from epsomite crusts, iron from red granite clay chelated by oak tannins, and micronutrients from local ores and sediments. You can either brew from these materials or buy the three-part Masterblend programme; both work, and the target profile is the same.

Target profile (per gallon): N 116 ppm · P 47 ppm · K 188 ppm · Ca 113 ppm · Mg 41 ppm · S 58 ppm, plus Fe 2.4 · B 1.2 · Mn 1.2 · Zn 0.3 · Cu 0.3 · Mo 0.06 ppm.

- For the **purchased** three-part programme — Masterblend 4-18-38, calcium nitrate, Epsom salt — see **Guide E: Modern Masterblend Guide**.
- For the **locally-sourced** brewing recipes — oak-ash potassium, aerated guano tea, caliche-vinegar calcium, epsomite magnesium, tannin-chelated iron — see **Guide F: Hydroponic Nutrients Guide**.
- For a fruiting-crop example with a full week-by-week schedule, see **Guide A: Pepper Growing Guide** (Chapter 7) or **Guide B: Bato Bucket Hydroponic Setup** (Section 5).

5. pH — short version

The pH window is 5.8 – 6.2. Outside that range, even perfectly formulated nutrients lock out. Both pH down and pH up can be made from local materials: prickly-pear vinegar for gentle daily acidification, fermented citric acid for stronger work, sulphur-burn acid for heavy lifts; oak-ash lye for alkali (it adds potassium at the same time), or hydrated lime from caliche. Always dose drop by drop from a dilute stock solution, stir, and wait ten minutes before re-testing.

For full recipes — including quantities, safety notes, and how to tell when a batch is ready — see **Guide G: pH Up & pH Down Replacements**.

6. Seasonal Growing Calendar

The Dragoons have four distinct growing windows. Unlike low-desert Arizona, elevation gives you a true cool season and a manageable summer. Plan crops around these windows.

Season	Months	Conditions	Best crops	Key tasks
Spring	Mar – May	Warming, low humidity, mild nights	Tomatoes, peppers, cucumbers, basil	Start seedlings indoors. Prep monsoon catchment.
Monsoon	Jul – Sep	Hot days, humid nights, afternoon storms	Basil, herbs, beans, heat-tolerant greens	Collect rainwater. Watch for fungus. Reduce nutrients.

Season	Months	Conditions	Best crops	Key tasks
Fall ★ *Best*	Sep – Nov	Perfect temps, clear skies, low humidity	Everything — tomatoes, peppers, lettuce, kale, chard	Plant heavily. Use collected monsoon water.
Winter	Nov – Feb	Cool to cold, frost risk	Spinach, arugula, kale, mache, cilantro	Hoophouse on. Excellent cold-greens window.

Monthly action guide

Month	Key actions
January	Grow cold-tolerant greens under cover. Plan spring seed orders.
February	Start tomato and pepper seeds indoors. Clean and prep reservoirs.
March	Transplant tomatoes and peppers. Set shade cloth for summer.
April	Peak spring growing. Monitor pH and EC daily as temps rise.
May	Install shade structures. Insulate or bury reservoirs before the heat.
June	Reduce plantings. Focus on heat-tolerant herbs. Pre-monsoon prep.
July	Monsoon begins — start water collection. Watch for fungal issues.
August	Continue collection. Harvest summer crops. Start fall seeds.
September	Fall season begins — the best growing period. Plant everything.
October	Peak fall harvest. Plant winter greens. Fill storage tanks.
November	Frost protection in place. Transition to cold-season crops.
December	Winter greens under cover. Review the season, order seeds.

7. Crop Selection

Crop selection for the Dragoons favours heat tolerance in summer, cold tolerance in winter, and Southwest-adapted varieties year-round. Elevation widens your palette beyond what low-desert growers can manage.

Crop	Variety	Season	Notes
Cherry tomato	Solar Fire, Sungold, Heatmaster	Spring / Fall	Heat-tolerant; sets fruit even at 95 °F
Chile pepper	Hatch, Ancho, Chiltepin (native)	Spring – Fall	Chiltepin is native to Cochise County — thrives here
Bell pepper	Karma, Corno di Toro	Fall	Best in fall — avoids summer heat stress
Cucumber	Diva, Marketmore	Spring / Monsoon	Loves humidity — excellent during monsoon
Kale	Lacinato, Red Russian	Fall / Winter	Sweeter after frost — top cold-season crop
Spinach	Bloomsdale, Space	Winter	Cool-weather star — ideal hoophouse crop
Arugula	Wild or cultivated	Fall / Winter / Spring	Fast (30 days), bolt-resistant in cool weather
Lettuce	Oak Leaf, Butterhead, Romaine	Fall / Winter / Spring	Avoid summer — bolts above 80 °F
Basil	Genovese, Thai, Holy	Summer / Monsoon	Loves the heat and monsoon humidity
Cilantro	Slow-bolt varieties	Fall / Winter / Spring	Bolts in summer — cool-season only
Oregano	Greek, Mexican	Year-round	Drought-tolerant, thrives at altitude
Epazote	Native variety	Summer / Fall	Traditional SW herb, native, heat-tolerant

Chiltepin is native to Cochise County and the surrounding sky islands. It is extraordinarily well adapted to your exact conditions — consider it your signature local crop.

For Andean peppers specifically — Aji, Rocoto, Manzano and why the Dragoons' elevation suits them — see **Guide A: Pepper Growing Guide**. For wine and table grapes in the same climate, see **Guide H: Grape Growing Guide**.

8. Troubleshooting

Most hydroponic problems in this region fall into three categories: pH drift from hard-water bicarbonates, heat-related root stress, and nutrient deficiencies from imprecisely sourced local inputs.

Symptom	Likely cause	Local fix
Yellow leaves between green veins	Iron deficiency (most common)	Brew fresh oak tannin + red clay chelated iron; foliar first for speed
pH rising constantly	Bicarbonates in well water	Pre-treat water with prickly-pear vinegar before nutrients; full bicarbonate scrub
Stunted growth, dark green leaves	Phosphorus deficiency	Stronger aerated bat-guano tea; ensure aeration is vigorous enough for nitrification
Brown leaf edges, tip burn	Calcium deficiency or salt burn	More caliche-vinegar extract; check EC — hard water may have pushed it too high
Wilting despite wet roots	Root rot from warm reservoir	Insulate or bury reservoir; add airstone; keep root zone below 72 °F
White crusty deposits on media	Mineral build-up from hard water	Flush with clean rainwater; switch to monsoon water as primary source
Powdery white coating on leaves	Powdery mildew — monsoon humidity	Increase airflow; dilute apple-cider vinegar spray; remove affected leaves
Slow growth in summer	Heat stress above 95 °F	30–40 % shade cloth; bury reservoir; pause fruiting crops; switch to heat-tolerant herbs
Nutrient solution turns cloudy	Bacterial growth from organic inputs	Replace reservoir; filter teas more finely; use within 48 hours of brewing

Essential tools

Tool	Why it matters
EC/TDS meter	Non-negotiable — know your water and nutrient concentration
pH meter	Non-negotiable — the most important daily measurement
Thermometer	Monitor reservoir and air temps — root zone must stay below 72 °F
Fine-mesh strainer / cloth	Filter all locally-brewed teas before they enter the reservoir
Dark covered containers	For reservoirs — prevent algae and UV degradation
Shade cloth (30–40 %)	Essential for summer UV protection
Rainwater catchment	Cistern + gutters for monsoon collection — your best water source
pH calibration solution	Keep the meter accurate — recalibrate monthly

Closing

The Dragoon Mountains are an exceptional environment for hydroponic growing. The monsoon gives you soft water; the elevation gives you cool summers and a true winter; the local geology gives you every element you need; and the long, clear fall gives you the best single growing window in the Southwest.

This primer is deliberately thin. It is the map, not the terrain. When you want to build a system, brew a nutrient, tune an acid, or understand why one pepper sets fruit at 95 °F and another does not, step sideways into the companion guide that covers it. The library is designed that way — shallow here, deep where it matters.

Plant for fall. Collect the monsoon. Treat the water. The rest follows.